

SECOND MID TERM EXAMINATION
B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Second Semester, April, 2026

Paper Code: BS 106

Subject: Applied Physics-II

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
Question - 1			
1 (a)	The powder of BCC structured crystal is studied with X-rays of wavelength 2 Å . The (210) reflection is observed at Bragg's angle 35°. Calculate the lattice parameter.	2.5	3
1 (b)	Prove that c/a ratio for HCP Lattice is $\sqrt{8/3}$.	2.5	3
1 (c)	Explain why a semiconductor acts as an insulator at 0 K.	2.5	4
1 (d)	In an intrinsic semiconductor ($E_g = 0.676 \text{ eV}$), $m_e = 0.09 m$ and $m_h = 0.36 m$. Calculate the concentration of intrinsic charge carriers at 300 K.	2.5	4
Question - 2			
2 (a)	Find the Miller indices of a plane that makes an intercept of 3 Å, 4 Å, and 5 Å on the coordinate axes of an orthorhombic crystal with a : b : c = 1 : 2 : 5 . Find the equation of the plane.	5	3
2 (b)	Define Point defect and show that the concentration of Schottky defects increases with temperature.	5	3
Question - 3			
3 (a)	Explain Kronig-Penney (KP) model for the motion of an electron in a periodic potential.	7	4
3 (b)	What do you understand by effective mass of an electron? Explain its significance.	3	4
Question - 4			
4 (a)	Draw sketch illustrating (101) and (222) planes. Also, show that diamond is a loosely packed structure.	5	3
4 (b)	Describe the doping features of a tunnel diode. Plot its I-V characteristics and explain the nature of the curve.	3	4
4 (c)	Explain how an LED works.	2	4

**** End of Question Paper ****